Everseen Vision AI Hackathon

Solving Real-World Retail Challenges with Computer Vision

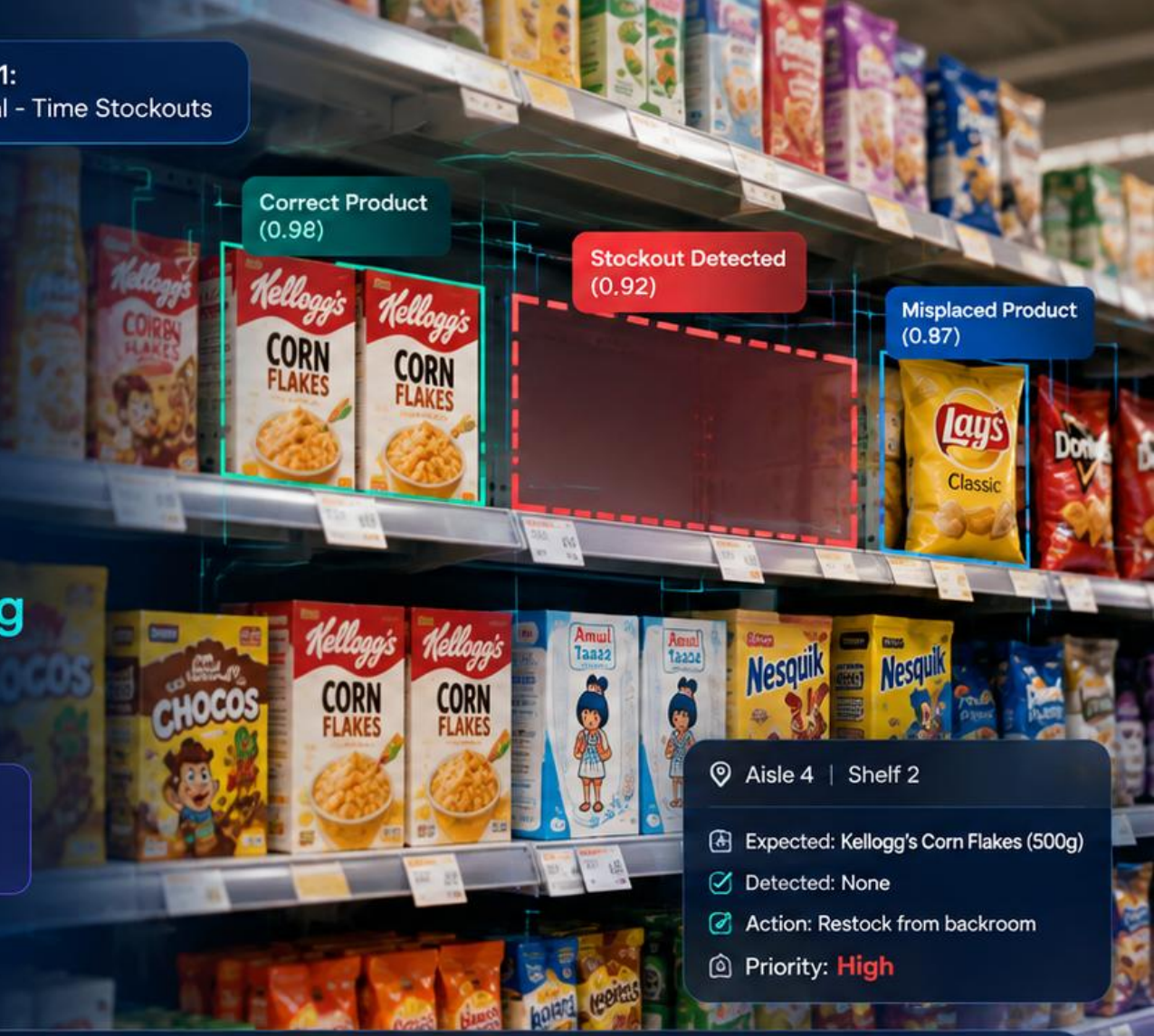
Problem Statement 1:
Shelf Auditing and Real - Time Stockouts

OptiShelf Temporal AI

Intelligent, Uncertainty-Aware Shelf Auditing

See the gap. **Verify** the evidence. **Trigger** the right action.

- PLANOGRAM INTELLIGENCE
- TEMPORAL VERIFICATION
- ACTIONABLE ALERTS



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A missing product is easy to see.
A reliable stockout is harder to confirm.

The Retail Problem

Manual audits and basic vision systems fail to provide reliable, real-time shelf intelligence.



Shelf Gaps

Products are unavailable in their designated locations.



Misplaced Products

Items occupy the wrong planogram positions.



False Alerts

Temporary occlusions and poor visual evidence can trigger unnecessary staff interventions.

Our Opportunity

Move beyond simple product counting to a smarter, more reliable shelf understanding and decision system.

Conventional Approach		OptiShelf Approach
Single-frame detection	→	Evidence across observations
Basic product counting	→	Planogram-aware comparison
Immediate, noisy alerts	→	Confidence-aware, prioritized actions

Key Takeaway

Retailers need a system that **understands** what belongs on a shelf, **verifies** whether an anomaly is persistent, and **recommends the next action**.

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Problem Statement 1: Shelf Auditing and Real - Time Stockouts

OptiShelf Temporal AI

From raw shelf imagery to verified retail decisions.

A four-stage intelligence pipeline that combines **spatial reasoning**, **temporal verification** and **actionable prioritization**.

01 Visual Perception

Analyze shelf images or video frames to localize visible products and shelf regions.



Detected products and shelf structure

02 Planogram Intelligence

Compare the observed product arrangement against the expected digital shelf layout using spatial and sequence alignment.



Planogram vs observed arrangement

03 Temporal Evidence Fusion

Track anomalies across repeated observations and defer definitive alerts when visibility is obstructed or evidence is uncertain.



Verify persistence across multiple frames

04 Decision and Action

Prioritize persistent anomalies and generate evidence-backed restocking, displacement, or verification tasks.

Restock Alert — High Priority

Kellogg's Corn Flakes (500g)

- Aisle 4 | Shelf 2
- Expected: 4 facings
- Detected: 0 facings
- Action: Restock from backroom

Actionable, prioritized task for staff

Proposed Technical Core

- **Pretrained object detection** (e.g., YOLO / DINO)
- **Visual feature embeddings** for product matching
- **Modified Needleman-Wunsch sequence alignment** for planogram comparison
- **Temporal state machine** for anomaly verification
- **Confidence- and severity-aware alert logic**

Our Key Differentiator

Spatial reasoning + Temporal verification + Actionable prioritization

Unlike a basic product counter, OptiShelf considers both where a product should be and whether the apparent discrepancy persists over time.

Store Camera Feed → OptiShelf AI Engine → Verified Anomalies → Store Staff Action

Built for practical deployment. Designed to prove its value.

System Architecture
From camera feeds to prioritized store actions

Shelf imagery
Camera / video



- Existing CCTV / store cameras
- Aisle and shelf metadata
- Live or recorded video frames

Vision engine
Detection / embeddings



- Product and shelf detection
- Visual feature embeddings
- Shelf region segmentation

Planogram + time
Alignment / verification



- Compare with digital planogram
- Sequence alignment (Needleman–Wunsch)
- Temporal evidence fusion

Prioritized anomaly and action engine
Restock · Investigate · Recheck · Resolve



- Confidence- and severity-aware prioritization
- Generate evidence-backed tasks
- Integrate optional sales-velocity metadata

Key Innovations
What makes OptiShelf different

1 **Planogram-aware spatial reasoning**
Detects discrepancies in expected product order and shelf placement, not just changes in object counts.

2 **Temporal uncertainty management**
Distinguishes persistent anomalies from temporary visibility problems through repeated observations.

3 **Operational prioritization**
Ranks tasks using anomaly severity, visual confidence, and optional sales-velocity information.

Prototype Validation Plan
Test on diverse shelf scenarios



Correctly stocked shelf
No alert expected



Persistent product removal
Stockout alert



Temporary visual obstruction
No immediate alert



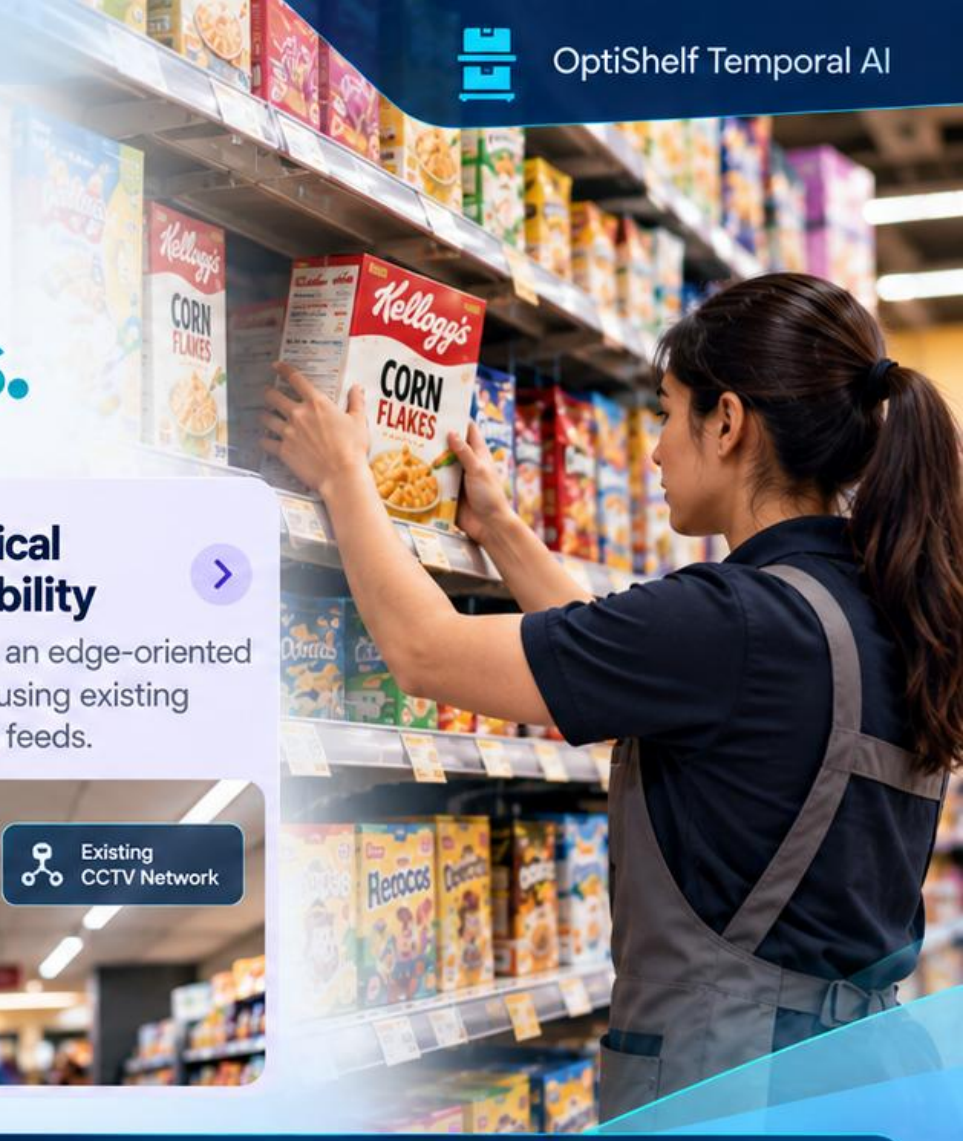
Product displacement
Misplacement alert

Evaluation Metrics
Measure performance and reliability

- Stockout-event precision and recall
- Transient false-alert rate
- Planogram mismatch detection
- Time to verified alert

Prototype results will be measured experimentally; production-scale performance remains to be validated.

Better shelf intelligence.
More confident store decisions.



Fewer false alerts

Verify visual evidence before escalating anomalies.



Faster intervention

Direct staff attention toward persistent shelf issues.



Practical scalability

Explore an edge-oriented design using existing camera feeds.



OptiShelf Temporal AI

Not just detecting what is missing — **understanding what needs attention.**

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Thank You!